Proposed Methodology for Deepfake Audio Detection

ASVspoof 2019 LA Dataset

- Real speech: Bona fide
- Fake speech: Spoof
- Known and unknown attacks

Preprocessing

- Mono conversion
- 16 kHz resampling
- Waveform normalization
- Variable-length handling
- Padding and masking

Training Augmentation

- RawBoost ① + ②
- Linear and non-linear distortion
- Impulsive signal-dependent noise
- Codec augmentation

Teacher Model

- Wav2Vec2-XLSR front-end
- AASIST back-end
- P2SGrad / OC-Softmax loss

Teacher Soft Predictions

- Class confidence scores
- Soft labels for student training

Student Model

- Frozen Wav2Vec2-XLSR front-end
- AASIST-L lightweight back-end

Knowledge Distillation Training

- Hard label loss
- Knowledge distillation loss
- Teacher knowledge transfer

Final Prediction

- Bona fide
- Spoof

Evaluation

- EER
- min-tDCF
- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix